

RGB Mood Light PCB

Overview

A USB-powered RGB mood light designed in KiCad as a beginner-to-intermediate PCB design project.

Features

• Arduino Nano ESP32 controller • USB-C power input • 6 WS2812B addressable RGB LEDs • Push-button user control • Per-LED 0.1 μ F decoupling capacitors • Ground plane implementation • ERC clean schematic • DRC clean PCB

Tools Used

• KiCad • Arduino Nano ESP32 • WS2812B LEDs

Design Process

1. Created schematic and organized into functional blocks.
2. Assigned footprints.
3. Performed PCB component placement.
4. Routed signal and power traces.
5. Added ground plane.
6. Fixed ERC/DRC violations.
7. Verified final design.

Skills Demonstrated

• Schematic Capture • PCB Layout • Routing • Power Distribution • Design Rule Checking • Hardware Debugging